

Assignment #3 → NETS, SME, Augmented Reality

The first speakers from this lecture were members of NETS. The NETS president and officers led the pitch and advertised the society as a place to build skills and connections outside of the classroom. They run labs, workshops, company tours, and guest speaker sessions. Their biggest event is their Cloudathon that is held each spring. They also do social events such as pumpkin carving and group outings, such as the NETS at Dough Zone event. Members can reach them at sjsu.nets@gmail.com, sjsunets.org, LinkedIn, and Instagram, and the club room in IS-117B when it is open.

The second batch of speakers were from SME (Society of Manufacturing Engineers). The president, treasurer, secretary, and social operations lead described the society as a bridge between the classroom concepts and the industry tasks. Members get industry exposure, hands-on projects, and professional connections. The club has one or two events a month, including tours such as Rigetti Quantum Computing. They also team with other clubs and societies to build projects together.

The main speaker of the day was our professor. He spoke on augmented reality within manufacturing. The presentation was filled with facts, statistics, and other data that he used to drive the entire presentation. Citing Deloitte research, he stated the following:

- About 2.4 million manufacturing positions will go unfilled from 2024 to 2040 as workers retire and automation pushes other workers out of the field.
- Mistakes and defects make up 25% of manufacturing costs, which adds up to 25 billion dollars each year when inspection and testing costs are included.
- By 2030, 75% of capital-equipment-intensive industries are expected to use AR as a key tool for cost reduction among frontline workers

He then went on to state and display how AR was being used within the industry already through AR applications on headsets, iPads, iPhones, etc. Some of the biggest use cases at this time are using AR within recruitment and interviews, technical training, work instructions, remote assistance in real-time, and knowledge capture from the retiring workforce.

The biggest takeaway for me was how my major, CNSM, can still get me into the manufacturing workforce due to the emerging developments in IoT among vehicles. Many CNSM graduates are being hired into manufacturing to connect machines, sensors, and AR devices that all depend on the network around them being reliable and secure. As automation continues to grow, there also need to be workers to manage the automation machines and ensure their connectivity and functionality. The numbers on AR showed me that new technology gets adopted when it ties directly into cost, quality, and safety of workers and industry operations. The exponential growth of new technologies shows that now is a good time to join the society workshops and grow my connections to hopefully be a part of the emerging development and implementation of augmented reality.